

4 Kinetic Energy

The total kinetic energy of the N -particle system is defined by the sum of the individual kinetic energies:

$$T = \sum_{\alpha=1}^N T_{\alpha} = \sum_{\alpha=1}^N \frac{1}{2} m_{\alpha} \vec{v}_{\alpha} \cdot \vec{v}_{\alpha} = \frac{1}{2} \sum_{\alpha=1}^N m_{\alpha} \dot{\vec{r}}_{\alpha}^2. \quad (40)$$

We now express every velocity relative to the centre of mass. From the position decomposition

$$\vec{r}_{\alpha} = \vec{R} + \vec{r}'_{\alpha} \quad (41)$$

it follows at once that

$$\vec{v}_{\alpha} = \dot{\vec{r}}_{\alpha} = \dot{\vec{R}} + \dot{\vec{r}}'_{\alpha}. \quad (42)$$

Substitute this decomposition into the kinetic-energy sum:

$$T = \frac{1}{2} \sum_{\alpha=1}^N m_{\alpha} (\dot{\vec{R}} + \dot{\vec{r}}'_{\alpha}) \cdot (\dot{\vec{R}} + \dot{\vec{r}}'_{\alpha}). \quad (43)$$

Expand the scalar product:

$$(\dot{\vec{R}} + \dot{\vec{r}}'_{\alpha}) \cdot (\dot{\vec{R}} + \dot{\vec{r}}'_{\alpha}) = \dot{\vec{R}} \cdot \dot{\vec{R}} + 2\dot{\vec{R}} \cdot \dot{\vec{r}}'_{\alpha} + \dot{\vec{r}}'_{\alpha} \cdot \dot{\vec{r}}'_{\alpha}. \quad (44)$$

Insertion of the expanded product yields three separate sums:

$$\begin{aligned} T &= \frac{1}{2} \sum_{\alpha} m_{\alpha} (\dot{\vec{R}} \cdot \dot{\vec{R}}) + \sum_{\alpha} m_{\alpha} (\dot{\vec{R}} \cdot \dot{\vec{r}}'_{\alpha}) + \frac{1}{2} \sum_{\alpha} m_{\alpha} (\dot{\vec{r}}'_{\alpha} \cdot \dot{\vec{r}}'_{\alpha}) \\ &= \frac{1}{2} \left(\sum_{\alpha} m_{\alpha} \right) \dot{\vec{R}}^2 + \dot{\vec{R}} \cdot \left(\sum_{\alpha} m_{\alpha} \dot{\vec{r}}'_{\alpha} \right) + \frac{1}{2} \sum_{\alpha} m_{\alpha} \dot{\vec{r}}'_{\alpha}{}^2. \end{aligned} \quad (45)$$

The coefficient of the first term is simply the total mass M . The coefficient of the second term vanishes by the centre-of-mass condition.

The kinetic energy consequently separates into two independent contributions:

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \sum_{\alpha=1}^N m_{\alpha} \dot{\vec{r}}'_{\alpha}{}^2. \quad (46)$$

The first term is the kinetic energy that would be possessed by a single particle of mass M moving with the centre-of-mass velocity. The second term is the kinetic energy associated with the motion of the particles relative to the centre of mass (the “internal” kinetic energy).

5 Conservation of Energy

We begin from the single-particle work–energy theorem. For each particle α the infinitesimal change in its kinetic energy equals the work done on it by the net force acting on it:

$$dT_{\alpha} = \vec{F}_{\alpha} \cdot d\vec{r}_{\alpha}. \quad (47)$$

Summing over all particles and integrating from an initial configuration (labelled “1”) to a final configuration (labelled “2”) yields the global statement

$$\Delta T = T_2 - T_1 = \sum_{\alpha=1}^N \int_1^2 \vec{F}_{\alpha} \cdot d\vec{r}_{\alpha}. \quad (48)$$

(The left-hand side is simply the sum of the individual changes ΔT_{α} .)

5.1 Separation of external and internal work

Decompose the force on particle α into its external and internal parts:

$$\vec{F}_\alpha = \vec{F}_\alpha^{\text{ext}} + \sum_{\substack{\beta=1 \\ \beta \neq \alpha}}^N \vec{F}_{\alpha\beta}. \quad (49)$$

Substitution into the work integral produces two distinct contributions:

$$\Delta T = W^{\text{ext}} + W^{\text{int}}, \quad (50)$$

where the external work is

$$W^{\text{ext}} = \sum_{\alpha=1}^N \int_1^2 \vec{F}_\alpha^{\text{ext}} \cdot d\vec{r}_\alpha \quad (51)$$

and the internal work is

$$W^{\text{int}} = \sum_{\alpha=1}^N \sum_{\substack{\beta=1 \\ \beta \neq \alpha}}^N \int_1^2 \vec{F}_{\alpha\beta} \cdot d\vec{r}_\alpha. \quad (52)$$

5.2 External work expressed as a change in potential

Assume every external force is conservative. Then there exists a potential-energy function $U^{\text{ext}}(\vec{r}_1, \dots, \vec{r}_N)$ such that

$$\vec{F}_\alpha^{\text{ext}} = -\nabla_\alpha U^{\text{ext}}, \quad (53)$$

where ∇_α denotes the gradient with respect to the coordinates of particle α . The external work therefore becomes a perfect differential:

$$W^{\text{ext}} = \sum_\alpha \int_1^2 (-\nabla_\alpha U^{\text{ext}}) \cdot d\vec{r}_\alpha = - \int_1^2 dU^{\text{ext}} = -(U_2^{\text{ext}} - U_1^{\text{ext}}) = -\Delta U^{\text{ext}}. \quad (54)$$

5.3 Internal work expressed as a change in potential

The internal-work double sum can be rewritten by pairing the mutual interaction of each pair of particles. Because $\vec{F}_{\beta\alpha} = -\vec{F}_{\alpha\beta}$ we have

$$\begin{aligned} W^{\text{int}} &= \sum_\alpha \sum_{\beta \neq \alpha} \int_1^2 \vec{F}_{\alpha\beta} \cdot d\vec{r}_\alpha \\ &= \frac{1}{2} \sum_\alpha \sum_{\beta \neq \alpha} \int_1^2 (\vec{F}_{\alpha\beta} \cdot d\vec{r}_\alpha + \vec{F}_{\beta\alpha} \cdot d\vec{r}_\beta) \\ &= \frac{1}{2} \sum_\alpha \sum_{\beta \neq \alpha} \int_1^2 \vec{F}_{\alpha\beta} \cdot (d\vec{r}_\alpha - d\vec{r}_\beta) \\ &= \sum_{\alpha < \beta} \int_1^2 \vec{F}_{\alpha\beta} \cdot d\vec{r}_{\alpha\beta}, \end{aligned} \quad (55)$$

where we have introduced the relative vector

$$\vec{r}_{\alpha\beta} = \vec{r}_\alpha - \vec{r}_\beta \quad (56)$$

and the notation $\sum_{\alpha < \beta}$ means that each unordered pair is counted once.

If the internal forces are themselves conservative (and therefore central), each pair interaction derives from a potential that depends only on the separation:

$$\vec{F}_{\alpha\beta} = -\nabla_{\alpha\beta} U_{\alpha\beta}(r_{\alpha\beta}). \quad (57)$$

The internal work then collapses to a pure difference of potential energies:

$$W^{\text{int}} = \sum_{\alpha < \beta} \int_1^2 (-\nabla_{\alpha\beta} U_{\alpha\beta}) \cdot d\vec{r}_{\alpha\beta} = - \sum_{\alpha < \beta} (U_{\alpha\beta}(2) - U_{\alpha\beta}(1)) = -\Delta U^{\text{int}}, \quad (58)$$

where the total internal potential is defined by

$$U^{\text{int}} = \sum_{\alpha < \beta} U_{\alpha\beta}(r_{\alpha\beta}). \quad (59)$$

5.4 Mechanical-energy conservation

Combining the external and internal results we obtain

$$\Delta T = -\Delta U^{\text{ext}} - \Delta U^{\text{int}}. \quad (60)$$

Defining the total potential energy by

$$U = U^{\text{ext}} + U^{\text{int}} \quad (61)$$

and the total mechanical energy by

$$E = T + U, \quad (62)$$

we arrive at the conservation statement

$$\Delta E = 0 \quad \text{or} \quad E = \text{constant}. \quad (63)$$

Thus, whenever every force that acts on the particles (both external and internal) is conservative, the total mechanical energy of the system is a constant of the motion.

When non-conservative forces are also present their work appears as an additional term on the right-hand side:

$$\Delta E = W^{\text{nc}}. \quad (64)$$

In the frequent case of kinetic friction the non-conservative work is negative, so the mechanical energy of the system decreases.

6 Summary

- The total linear momentum of an N -particle system changes only under the action of the net external force. Internal forces that obey Newton's third law cancel in pairs.
- The centre of mass moves exactly as a single particle of mass M subject to the net external force.
- The total angular momentum changes only under the action of the net external torque, provided all internal forces are central.
- Kinetic energy separates into a centre-of-mass term and a relative-motion term.
- When all forces are conservative the total mechanical energy $T + U$ is conserved.

Lagrangian Formulation — Generalized Coordinates

Classical Mechanics — Lecture Notes

1 Generalized Coordinates

Generalized coordinates are independent variables that uniquely specifies the configuration of a mechanical system.

From the preceding sections we recall that the position of a particle may be written as the Cartesian vector

$$\vec{r} = (x, y, z) \quad (1)$$

or, equally well, in spherical polar coordinates (r, θ, ϕ) or cylindrical coordinates (ρ, ϕ, z) . Any particular mechanical problem can be formulated in terms of many different sets of coordinates.

To specify the position of a single particle in three-dimensional space we need three coordinates: (x, y, z) . For two particles we need six coordinates

$$(x_1, y_1, z_1, x_2, y_2, z_2). \quad (2)$$

For three particles we need nine coordinates, and in general for N particles we need $3N$ coordinates

$$(x_1, y_1, z_1, \dots, x_N, y_N, z_N). \quad (3)$$

As we know, some problems are solved more easily in one coordinate system and some are solved more easily in another. To avoid being specific about the coordinate system we introduce a new set of coordinates called *generalized coordinates*. In this language the position and velocity of a particle are denoted by q_i and $\dot{q}_i$ respectively.

1.1 Transformation Equations

For a specific problem involving one particle we choose generalized coordinate system. We write collectively

$$q_i = (q_1, q_2, q_3). \quad (4)$$

The generalized coordinates are defined as functions of the Cartesian coordinates and of time through a transformation relation:

$$\begin{aligned} q_1 &= q_1(x, y, z, t), \\ q_2 &= q_2(x, y, z, t), \\ q_3 &= q_3(x, y, z, t). \end{aligned} \quad (5)$$

Similarly, for an N -particle system the analogous statements read

$$\begin{aligned} q_1 &= q_1(x_1, y_1, z_1, \dots, x_N, y_N, z_N, t), \\ q_2 &= q_2(x_1, y_1, z_1, \dots, x_N, y_N, z_N, t), \\ &\vdots \\ q_{3N} &= q_{3N}(x_1, y_1, z_1, \dots, x_N, y_N, z_N, t). \end{aligned} \quad (6)$$

Each generalized coordinate depends on all of the Cartesian coordinates and on t through equation (6).

The inverse transformation relations for an N -particle system are

$$\begin{aligned} x_1 &= x_1(q_1, q_2, \dots, q_{3N}, t), \\ y_1 &= y_1(q_1, q_2, \dots, q_{3N}, t), \\ &\vdots \\ z_N &= z_N(q_1, q_2, \dots, q_{3N}, t). \end{aligned} \quad (7)$$

Each Cartesian coordinate depends on all of the generalized coordinates and on time through equation (7).

It is often convenient to denote the whole set of Cartesian coordinates by the single letter x . For a single particle we simply relabel

$$(x, y, z) \longrightarrow (x_1, x_2, x_3). \quad (8)$$

The same convention extends immediately to N particles:

$$\begin{cases} x_1 &= x_1(q_1, q_2, \dots, q_n, t), \\ x_3 &= x_2(q_1, q_2, \dots, q_n, t), \\ &\vdots \\ x_n &= x_n(q_1, q_2, \dots, q_n, t), \end{cases} \quad (9)$$

where $n = 3N$, and

$$\begin{cases} q_1 &= q_1(x_1, x_2, x_3, \dots, x_n, t), \\ q_2 &= q_2(x_1, x_2, x_3, \dots, x_n, t), \\ &\vdots \\ q_n &= q_n(x_1, x_2, x_3, \dots, x_n, t). \end{cases} \quad (10)$$

1.2 Remark

The generalized coordinates q_i , unlike the Cartesian coordinates, do not in general divide into convenient groups of three that can be associated together to form ordinary vectors. Consequently we write the transformation relations equation (9) only in component form:

$$\begin{cases} x_i = x_i(\{q_j\}, t), \\ i = 1, 2, \dots, n, \\ j = 1, 2, \dots, n, \end{cases} \quad (11)$$

and

$$\begin{cases} \vec{r}_i = \vec{r}_i(\{q_j\}, t), \\ i = 1, 2, \dots, N, \\ j = 1, 2, \dots, n. \end{cases} \quad (12)$$

The position vector of the i -th particle retains its usual Cartesian expression

$$\vec{r}_i = x_i \hat{i} + y_i \hat{j} + z_i \hat{k}. \quad (13)$$

Lagrangian Formulation — Generalized Coordinates

Classical Mechanics — Lecture Notes

1 Degrees of Freedom and Constraints

Every physical system possesses a characteristic number of **degrees of freedom** (DoF). We define this quantity as the minimum number of independent coordinates required to uniquely specify the exact configuration of every particle in the system at any given moment.

1.1 Free Particles

- **Single Free Particle:** A lone particle moving unrestricted in three-dimensional physical space requires three independent coordinates to pinpoint its position. We can specify this location using Cartesian coordinates (x, y, z) , spherical coordinates (r, θ, ϕ) , or cylindrical coordinates (ρ, ϕ, z) . Therefore, its total degrees of freedom equal exactly 3:

$$\text{DoF} = 3 \tag{1}$$

- **System of N Free Particles:** If a mechanical system comprises N entirely unconstrained, independent particles, each individual particle possesses 3 independent spatial degrees of freedom. Consequently, the entire system requires $3N$ independent coordinates to locate every point simultaneously:

$$\text{DoF} = 3N \tag{2}$$

1.2 Constrained Motion

When a physical system faces geometric restrictions or mechanical bounds, we say the system is acted upon by forces of constraint. These physical restrictions introduce **mathematical dependencies among the coordinates**, allowing us to **reduce** the total number of parameters needed to describe the system's overall motion. As a result, constraints systematically decrease the degrees of freedom available to the system.

1.2.1 Example: A Particle Rotating in a Plane

Consider a single particle restricted to a flat two-dimensional plane (xy -plane), anchored to a central origin by a rigid, inextensible rod of fixed length r .

$$z = \text{constant} \quad (\text{Restricted to a 2D plane}) \tag{3}$$

$$r = \text{fixed length } l \quad (\text{Rigid constraint}) \tag{4}$$

Mathematically, if we begin with a 3D description (x, y, z) , we apply two independent constraint equations ($z = c$ and $x^2 + y^2 = l^2$). Thus, the system reduces from 3 original coordinates down to a single remaining independent coordinate, namely the angular parameter ϕ . The system possesses exactly one degree of freedom.

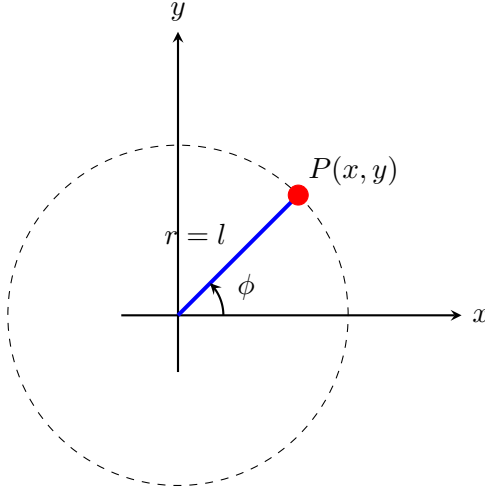


Figure 1: A particle confined to rotate along a circular pathway of fixed radius r in a plane.

1.3 Additional Geometry-Driven Constraint Examples

1.3.1 Unconstrained Planar Motion

If a particle moves freely within a two-dimensional flat plane without radial boundaries, we freeze its vertical component ($z = \text{constant}$).

$$\text{Number of Constraints } (k) = 1 \quad (5)$$

The original 3D system loses one degree of freedom, meaning that a set of two coordinates—either Cartesian (x, y) or polar (r, ϕ) —completely describes the particle's state:

$$\text{DoF} = 3 - 1 = 2 \quad (6)$$

1.3.2 The Plane Pendulum

Consider a simple (planar) pendulum oscillating under a rigid ceiling support line where $z = 0$. The particle hangs from a fixed pivot point via an unyielding string of length l , satisfying $x^2 + y^2 = l^2 = \text{constant}$.

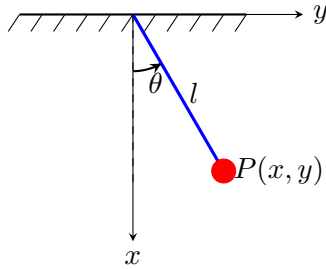


Figure 2: Coordinate structure of a single-link simple planar pendulum.

These combined limits constitute two explicit coordinate restrictions:

$$z = 0 \quad \text{and} \quad x^2 + y^2 = l^2 \quad (7)$$

We expand the local variables using trigonometry:

$$x = l \cos \theta, \quad y = l \sin \theta \quad (8)$$

Consequently, knowing the isolated angular displacement value $\theta = \tan^{-1}(y/x)$ is entirely sufficient to locate the oscillating particle P at any instant.